BiAgram Lab Notebook — Oct 18, 2026

Oct 18, 2026 · @J

Summary

Baselines 1 and 2 passed today, Qwen3-4B-Base became the project model, and test set v2.1 was built and frozen. The Stage 3 quick pass was running at 5:19 pm.

Goal for the session: get the healthy model to pass the ≥90% gate in both languages (Stage 1), reproduce Roll et al. (Stage 2), then start the lesion stages.

Key decisions

Decision	Reason
Project model = **Qwen/Qwen3-4B-Base**, float16	Only model tested that passed the Spanish gate (pre-set rule: best healthy Stage 1 + MultiBLiMP score). Apache 2.0, no license to sign.
Rejected Qwen2.5-3B-Instruct and Qwen2.5-3B (base)	Spanish gate 53–55%; verb-only test 69%; defaults to singular verb forms after plural subjects.
Stage 2 run on two models	Qwen2.5-3B-Instruct = calibration (Roll et al.'s own model, checks the code). Qwen3 = does the pattern carry over.
Test set v2.1, then **frozen**	Fixed design-rule violations found during the healthy check, before any lesion run on the test set.
Stage 3 quick pass adds layers 0, 1 and severity 0.75	Decided before any Stage 3 result (see Notebook changes).
Scoring metric unchanged (norm primary, sum and mean reported)	Qwen3 passes under all three; no more metric switching (stopping rule). Option A not implemented.

Status at end of session

- [x] ~~Stage 1 healthy baseline (Qwen3, v2.1): PASS~~

- [x] ~~Stage 2 Roll reproduction: pattern reproduced on Qwen2.5-3B-Instruct~~
- [x] ~~Stage 4 neuron selection computed and saved~~
- [ ] Stage 3 quick pass (running)
- [ ] Native Spanish speaker check of the lexicon
- [ ] ISEF research plan addendum for today's changes
- [ ] Stage 3 full sweep, Stage 4 lesion runs

Files

Four new repo files today; v2 and SPSF_03 are kept for the record.

File	Where	What it is	Status
`test_set/make_testset_v2_1.py`	repo	Generator for test set v2.1. Header lists every change from v2 and why. Refuses to run if a Spanish verb form is a common English word or a matched pair is broken.	new
`test_set/biagram_testset_v2_1.csv`	repo + Colab upload	958 pairs, 419 items. New columns `match_id`, `split`.	new, **frozen**
`notebooks/SPSF_04_stages.ipynb`	repo + Colab	Main notebook: Qwen3 defaults, v2.1, built-in false-positive checks, lesion check, updated Stage 3 sweep.	new

File	Where	What it is	Status
`README.md`	repo	Model choice, results table, run order, freeze rule.	updated
`test_set/make_testset_v2.py`, `biagram_testset_v2.csv`, `notebooks/SPSF_03_stages.ipynb`	repo	Previous version.	kept, not used
`stage1_healthy_2.csv`	Drive, BiAgram/results	Stage 1 scores, Qwen2.5-3B base on v2.	result
`stage1_healthy_Qwen3-4B-Base.csv`	Drive	Stage 1 scores, Qwen3 on v2.1.	result
`stage2_roll_Qwen3-4B-Base.csv`, `stage2_roll_Qwen2.5-3B-Instruct.csv`	Drive	Stage 2 lesion runs.	result
`stage4_activation_probs_Qwen3-4B-Base.pt`	Drive	p_en / p_es per neuron from 2,000 OPUS-100 en–es pairs.	result
`stage3_sweep_quick_Qwen3-4B-Base.csv`	Drive	Stage 3 quick pass, saved after each condition.	in progress

Check the exact Drive file names against this table; a Stage 2 run before the file-name fix may have saved as `stage2_roll_reproduction.csv`.

Log (in order)

The Spanish failure was traced to the model, not the code or the test set, after six steps. Times are approximate, Eastern.

Time	Step	Result	What it meant / next
afternoon	Stage 1 on **Qwen2.5-3B-Instruct**, float16, test set v2	EN 99.5% PASS, ES 53.1% FAIL	Spanish verb morphology near chance (~50%); articles 81–96%.
afternoon	Reran in float32	Same numbers	float16 was **not** the cause (my wrong guess). Kept float32 as safe for Qwen2.5.
afternoon	Fix A: **Qwen2.5-3B base** (MultiBLiMP reports post-training hurts)	EN 89.6% FAIL, ES 55.2% FAIL; MultiBLiMP 96.0 / 88.7	Base worse on every measure. Rejected; Instruct kept at that point.
4:09 pm	Subject check (does a plural subject push toward the plural verb?) on base	10 of 13 verbs (77%), with very large swings	Model uses the subject somewhat; noise from word splitting is large.
4:16 pm	Verb-only check (score only the verb given the words before it) on base	69%; plural subjects 25%	The model itself prefers *Los niños vende*. **Model limitation.** Ran fallback per stopping rule.
4:25 pm	Loaded **Qwen3-4B-Base**, float16	36 layers, 9,728 MLP neurons/layer, 8.0 GB; counting check OK (−0.08 / −0.13)	
4:30 pm	Stage 1 on Qwen3, v2	EN 97.9%, ES 96.4% PASS; subject 13/13; verb-only 98%	Passed every false-positive check. **Qwen3 selected.**
4:40 pm	Stage 2 on Qwen3	No drop anywhere (all < 1 pt)	Lesion check run to rule out a broken lesion.
4:42 pm	Lesion check on Qwen3	Zeros 0 → 0.75 → 0; L0 down 100% → 56.5%	Lesion code works; Qwen3 NO is real.

Time	Step	Result	What it meant / next
4:49 pm	Stage 2 on **Qwen2.5-3B-Instruct** (calibration)	Gate L7 −37.5; K −0.2; Up 3.3 → **YES**	Code reproduces Roll et al. Baseline 2 done.
~4:58 pm	Stage 4 neuron selection on Qwen3	1,751 neurons per set	Matches LAPE layer pattern.
~5:00 pm	Built test set v2.1 and SPSF_04	Passed toy-model tests	
5:14 pm	Stage 1 on Qwen3, **v2.1**	EN 99.0%, ES 97.9% PASS; subject 48/48	**v2.1 frozen.**
5:15 pm	Started Stage 3 quick pass	~17 s per condition, ~35 min for 126	Running.

Stage 1: healthy baseline

Qwen3-4B-Base on v2.1 passes the ⩾90% gate in both languages under all three scores; both Qwen2.5 models fail Spanish. Gate = % of items where the correct sentence beats every foil.

Run	EN gate	ES gate	ES gate: norm / sum / mean	MultiBLiMP eng / spa	ES subject check	ES verb-only	Code-switched (reported)
Qwen2.5-3B-Instruct, v2 (float16)	99.5% PASS	53.1% FAIL	53.1 / 47.9 / 59.9	97.0 / 90.3	not run	not run	40.0%
Qwen2.5-3B base, v2 (float32)	89.6% FAIL	55.2% FAIL	55.2 / 43.2 / 59.4	96.0 / 88.7	10/13 verbs	69%	28.6%
Qwen3-4B-Base, v2 (float16)	97.9% PASS	96.4% PASS	96.4 / 95.3 / 94.8	99.7 / 96.3	13/13 verbs	98%	31.4%
Qwen3-4B-Base, v2.1 (float16)	**99.0% PASS**	**97.9% PASS**	97.9 / 96.4 / 97.4	99.7 / 96.3	**48/48 matched pairs**	**99%**	37.1%

Qwen3 on v2.1 also passes the English checks: subject check 24/24, verb-only 100%, English gate 99.0 / 98.4 / 99.0.

Qwen3-4B-Base, v2.1, item accuracy by feature (%)

Feature	English	Spanish
Articles	100.0	100.0
Auxiliary	100.0	100.0
Tense	100.0	93.8
Verb agreement	95.8	97.9

Code-switched: determiner + noun 91.7%, auxiliary + verb 9.1%, pronoun pilot 8.3%. The healthy model prefers *La niña is cocinando* over the attested *La niña está cooking*, so code-switching cannot be a lesion measure yet.

Remaining Spanish misses on v2.1 (kept in; the set is frozen): *La maestra beber agua* preferred over *bebe* once (AGR-ES-014), and three tense items where the present beats the preterite even with *ayer* first (TNS-ES-016, -022, -032). English misses: AGR-EN-017 (*close*), AGR-EN-029 (*learning*).

Why these aren't false positives

1. The same test failed Qwen2.5 (53–55%), so the gate can fail; passing means something.
2. Subject check 48/48: a plural subject pushes toward the plural form for every matched pair. That comparison cancels form frequency and word splitting.
3. Verb-only (99%) ≈ whole sentence (100%): the words after the verb aren't doing the work.
4. Holds under all three scoring methods.
5. Independent outside test agrees (MultiBLiMP 96.3% Spanish).

Dialect check: Qwen3 preferred the Spain form in 3 of 4 pairs (*vosotros queréis*, *ordenador*, *conducir el coche*); Latin American only for *ustedes tienen razón*. The test set uses general Latin American Spanish. A note for the paper, not a fail.

Sanity (counting) check: Qwen3 −0.08 (es) / −0.13 (en); Qwen2.5-3B base −0.12 / −0.12; Qwen2.5-Instruct float32 −0.13 / −0.33. All OK (threshold −1).

Stage 2: Roll et al. reproduction

The lesion code reproduces Roll et al.'s pattern on their model (Gate at layer 7 drops BLiMP 37.5 points; K and Up don't), but Qwen3 shows no effect. Setup: 200 English BLiMP pairs (4 sets × 50), layers 7/18/28, K/Gate/Up, 75% of weights zeroed, 3 seeds each.

Target: Roll et al. Appendix S probed Qwen2.5-3B-Instruct on their own 70 items: Gate L7 −52.5 pp syntax, K and Up near baseline. Match = the same pattern, not the same size.

Lesion (75%)	Qwen2.5-3B-Instruct: mean BLiMP % (sd)	drop	Qwen3-4B-Base: mean BLiMP % (sd)	drop
Healthy	92.0	—	96.0	—
L7 Gate	54.5 (1.7)	**37.5**	96.7 (0.8)	−0.7
L7 K	92.2 (0.8)	−0.2	96.2 (0.8)	−0.2
L7 Up	88.7 (0.6)	3.3	97.0 (0.5)	−1.0
L18 Gate	91.8 (1.0)	0.2	95.8 (0.6)	0.2
L18 K	91.7 (0.6)	0.3	95.8 (1.2)	0.2
L18 Up	91.8 (1.5)	0.2	95.7 (0.3)	0.3
L28 Gate	90.8 (0.8)	1.2	95.5 (0.0)	0.5
L28 K	91.8 (0.3)	0.2	95.8 (0.3)	0.2
L28 Up	90.3 (0.6)	1.7	95.7 (0.8)	0.3
Pattern reproduced?	**YES**		**NO**	

Lesion check (was the damage real?)

Check	Qwen2.5-3B-Instruct	Qwen3-4B-Base
Fraction of L7 gate weights at zero: before / during / after	0.0 / 0.75 / 0.0	0.0 / 0.75 / 0.0
L7 Gate 100%	86.5%	96.0%
L0 Down 100% (positive control)	81.0%	**56.5%**

Check	Qwen2.5-3B-Instruct	Qwen3-4B-Base
L1 O 100%	**57.0%**	94.0%

Interpretation

- The calibration run checks the code: it gives Roll et al.'s result on Roll et al.'s model. The drop is smaller than theirs because the items differ (BLiMP vs. their 70).
- Qwen3 shrugs off single-layer damage at layers 7, 18 and 28, even with the whole layer-7 MLP silenced (Gate 100% → 96.0%). Roll et al. also saw no drop in a 7B model. Reported as "the pattern does not carry over to a newer model."
- **Removing a part hurts less than corrupting it.** Gate at 100% (MLP switched off cleanly) dropped Qwen2.5 to 86.5%; Gate at 75% (MLP still running on corrupted weights) dropped it to 54.5%. So 100% is not the most severe setting.
- Each model has a different weak spot: Qwen3 breaks when layer 0's MLP output is removed, Qwen2.5 when layer 1's attention output is.

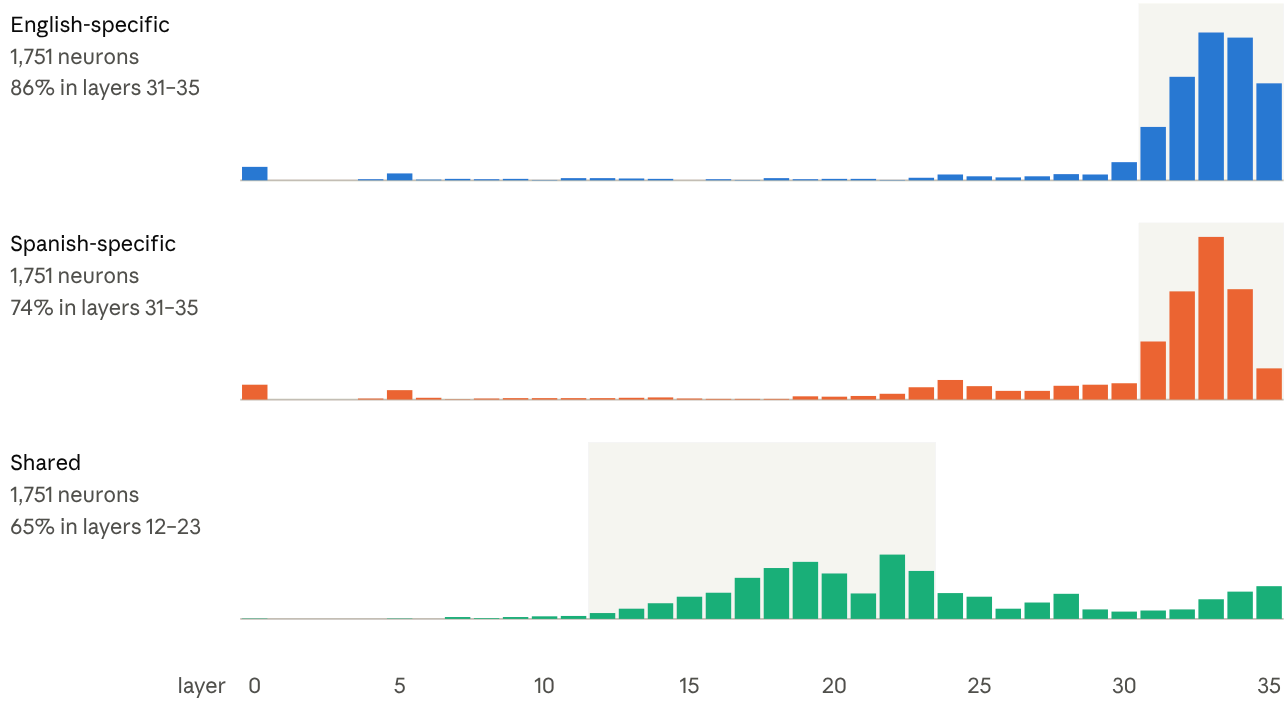
Stage 4: neuron selection

The selection behaves like the published method: language-specific neurons cluster in the last five layers (plus a small group at layer 0), shared neurons in the middle.

Method (fixed before any result): LAPE-style selection (Tang et al. 2024) on 2,000 OPUS-100 English–Spanish translation pairs, excluding test sentences. A neuron "fires" when its gate output is > 0. Each set = 0.5% of all MLP neurons (36 × 9,728 = 350,208 → 1,751), sets kept disjoint, `min_prob` = 0.2. Saved as `stage4_activation_probs_Qwen3-4B-Base.pt`.

Language-specific neurons cluster in the last layers, shared in the middle

Neurons chosen per layer (Qwen3-4B-Base, 0.5% rule), same scale in each panel. Shading = cluster zone.



Colab output, Stage 4 selection cell, Oct 18 2026 · 1,751 neurons per set, 36 layers

What to expect in Stage 4: the language-specific neurons sit where the model picks its output words. Silencing them may make the model drift into the other language rather than make grammar errors, as a 2026 study found on Llama (arXiv 2604.05090). Stage 4 measures drift, so either outcome is a result.

Test set v2.1

v2.1 fixes design-rule violations found during the healthy-model check and was frozen at 5:14 pm, before any lesion run on the test set. Same size as v2: 958 pairs, 419 items (192 English, 192 Spanish, 35 code-switched).

#	Change	Example	Why	Needed or extra?
1	Tense items put the time word **first**	*Ayer la niña arregló la bicicleta.* / *Yesterday the girl fixed the bike.* (was *… ayer.* at the end)	The model reads left to right, so with *ayer* last it can't use the cue when it scores the verb. All remaining v2 tense misses were this type. English tense went 95.8% → 100%.	Needed
2	Verb *barrer* replaced by *preparar* (*prepare the salad* / *preparar la ensalada*)	*barren* = 1 piece (the English word), *barre* = 2	A Spanish form that is an English word gets read as English. Caused a Qwen3 failure (*Las niñas barre*). Also removes *el piso*, a dialect-dependent word.	Needed
3	*vender* → *aprender* (*learn the lesson*); *contestar* → *estudiar* (*study the map*)	stem foils *vend*, *contest*	Broke the existing Design Rule that a stem foil must not be an English word. Should have been caught in v2.	Needed (rule violation)
4	*lavar* → *llevar* (*carry the bags* / *llevar las bolsas*)	*lava* is an English word	Same rule applied to every verb. Had **not** caused a failure.	Extra (precaution)

#	Change	Example	Why	Needed or extra?
5	Agreement and auxiliary items in **matched pairs** (column `match_id`)	*La niña cocina la cena* / *Las niñas cocinan la cena*	The subject check compares the same verb and object with a singular vs. plural subject, cancelling any built-in preference for one form.	Extra (cleaner check); items are newly drawn, so not one-to-one with v2
6	New column `split` = dev / locked, half the frames each, by feature		Build and debug Stage 4 on dev; look at locked once, for the final result. Not used until Stage 4.	Extra
7	Generator checks		Refuses to run if a Spanish form is in a common-English-word list, or a matched pair lacks a singular or plural item.	Extra (safety)

Tokenizer check behind changes 2–4: Qwen's tokenizer (rebuilt locally, matched the Colab splits exactly) splits *abre* = 1 piece vs. *abren* = 2, *cocina* 1 vs. *cocinan* 2, *barre* 2 vs. *barren* 1. Uneven piece counts tilt a pair before grammar comes in; the matched design (change 5) cancels that in the subject check.

Is changing items OK scientifically? Yes, under these conditions, which all hold:

- The reason is a rule written before the change, not "the model got it wrong."
- The rule was applied to every verb, including ones that never failed (*lava*).
- Failures that don't break a rule stayed in (*La maestra beber agua*, three tense items).
- No lesion results on the test set existed yet. This is **norming**: testing materials on a pilot (the healthy model) and repairing bad items before the real experiment.
- Every change is recorded (generator header, this page; v2 kept in the repo).

Gray area to state in the paper: *barren* was noticed because the model failed it. It was removed under the pre-existing rule, applied across the board.

Freeze rule: v2.1 is frozen as of Oct 18 2026. The only allowed change is a native Spanish speaker correcting real grammar errors, before Stage 3's full sweep. If a future check fails the gate, items are not edited; analysis is restricted to items the healthy model gets right, and the number dropped is reported.

Notebook changes (SPSF_03 → SPSF_04)

SPSF_04 = SPSF_03 plus today's fixes; the scoring code and lesion code are unchanged. 35 cells; passed the toy-model tests before delivery.

Area	Change	Why
Settings	`MODEL_NAME = "Qwen/Qwen3-4B-Base"`, `DTYPE = "float16"`, `TESTSET_PATH = "/content/biagram_testset_v2_1.csv"`, `BATCH_SIZE = 8`	Selected model and frozen test set. If the counting check fails, use `bfloat16`; float32 does not fit a 4B model on a T4.
Settings	New `MODEL_TAG` (model name) in every results file name	Without it the two Stage 2 runs overwrote each other.
Loading	`device_map=DEVICE, low_cpu_mem_usage=True` (already edited in Colab earlier)	Loading through regular memory first can crash free Colab.
Sanity check	Third sentence now *Ayer el niño aprendió la lección.*	*vender* is no longer in the test set. Cosmetic.
Stage 1	Built-in **subject check** and **verb-only check** cells	The false-positive checks run today, so a rerun gets the same evidence. Verb-only = targeted evaluation (Linzen et al. 2016; Marvin & Linzen 2018).
Stage 2	Corrected text: Roll et al. tested Qwen2.5-3B-Instruct only in Appendix S; run twice (calibration + project model)	The old text wrongly said "this exact model."

Area	Change	Why
Stage 2	New **lesion check** cell (zeros before/during/after, positive controls)	Rules out a lesion that silently does nothing.
Stage 3	Quick pass: layers **0, 1**, 3, 7, 18, 28 (layer 12 moved to the full sweep only); severities 0.5, **0.75**, 1.0; seed 0 → 126 conditions. Full sweep adds layers 0 and 1.	Decided before any Stage 3 result: Roll et al. found syntax symptoms peak in the earliest layers; Qwen3 ignored damage at 7/18/28; 75% hurt more than 100% in calibration. Cost: about double the quick-pass time.
Stage 4	Reuses `stage4_activation_probs_<model>.pt` if it exists	Selection depends only on the model and the OPUS text; no need to recompute.

Not changed, on purpose: scoring stays norm (sum / √tokens) as primary, sum and mean reported. Scoring "Option A" (sum for same-length pairs, norm for omissions) was not implemented. Qwen3 passes under all three, the lesion statistics use the per-token mean (which cancels length effects), and the stopping rule bans more metric switching.

Measured runtime (Qwen3, T4, float16): about 17 s per Stage 3 condition, so the quick pass (126) takes about 35 min and the full sweep (7 layers × 7 parts × 4 severities × 3 seeds = 588) about 3 hours.

Mistakes, open issues, next steps

Wrong turns today (recorded so they aren't repeated)

Wrong call	What showed it was wrong	Lesson
float16 was corrupting Qwen2.5's scores	float32 rerun gave nearly identical numbers	Test a suspected cause before switching settings.
A base model would fix Spanish (from MultiBLiMP's average finding)	Base scored worse on every measure	A finding averaged across many models may not hold for one model.

Wrong call	What showed it was wrong	Lesson
The subject-check cutoff (≥80% vs. ~50%) left a gap; 77% landed in it	Needed a second test (verb-only) to decide	Write decision rules with no gap.
First version of the subject-check cell crashed	`healthy` holds scores only, not sentences	Fixed by merging with `pairs`.
Stage 2 text said Roll et al. used "this exact model"	Rechecked the paper: main study used 1B models; Qwen2.5-3B was only an appendix probe	Re-read the source before citing a detail.

Open issues

- Code-switching: the healthy model doesn't follow code-switching constraints (aux + verb 9%, pronoun 8%). Reported only; can't be a lesion measure.
- Dialect: the model leans Spain Spanish (3 of 4); the test set is general Latin American. Note in the paper.
- Stage 3 interpretation: with 126 lesions some will match predictions by chance. Compare against Stage 4's random-neuron controls before reading anything into one lesion.
- Stage 4: language-specific neurons sit in the last layers, so silencing them may cause drift to the other language rather than grammar errors.
- Stage 2 Drive file names: check whether the first Qwen3 run saved under the old name `stage2_roll_reproduction.csv`.

Next steps

- [ ] Finish the Stage 3 quick pass and paste the summary table
- [ ] Commit `make_testset_v2_1.py`, `biagram_testset_v2_1.csv`, `SPSF_04_stages.ipynb`, `README.md` ("v2.1 test set frozen; Qwen3-4B-Base selected; Stages 1–2 done")
- [ ] ISEF research plan addendum: model change (Llama → Qwen2.5 → Qwen3-4B-Base), language pair (if the original plan said Portuguese), test set version; date and reason for each
- [ ] Native Spanish speaker check of the v2.1 lexicon (8 subjects, 14 verbs, 16 nouns) before the full sweep
- [ ] Update the Methodology doc to Qwen3-4B-Base

- [ ] Stage 3 full sweep (~3 hours, resumable)
- [ ] Stage 4 lesion runs (on dev first)
- Deadlines: core results Dec 10; ISEF submission Jan 9.

Sources

- Roll, Kries, Gwilliams & Shain 2026, Artificial Aphasias in Lesioned Language Models (arXiv 2605.16222): opened today for the models, layers, severities, Section 4.3 depth findings and Appendix S Qwen2.5-3B-Instruct probe.
- From the project literature review, not reopened today: Tang et al. 2024 (LAPE, ACL); Jumelet et al. 2026 (MultiBLiMP 1.0, TACL); Linzen, Dupoux & Goldberg 2016; Marvin & Linzen 2018; "Multilingual LMs Encode Script Over Linguistic Structure" (arXiv 2604.05090); BLiMP (Warstadt et al. 2020).